

general understanding of locomotives, battery chargers for telecommunications, inverter systems for applications involving renewable energy generation such as wind and photovoltaic conversion, and also general power utility systems may help you comprehend any unknown power system that you may come across.

Know the elements of power electronics

Undoubtedly, one of the things that makes power electronics interesting is its incorporation of elemental concepts from diverse set of fields. As a power electronics professional, you may come across the basics of analogue circuits, electronic devices, control systems, power systems, magnetics, electric machines and even numerical solutions.

Thus the practice of power electronics requires a broad understanding of electrical as well as electronics engineering. In addition, there are fundamental concepts which are unique to power electronics and require specialised study.

Let's start with the switching mode. High-frequency switching makes converter modeling the central element of your study. You need to know how to put a converter in equilibrium. The principles of steady-state converter analysis including inductor-volt-second balance, capacitor charge balance and small ripple approximation are of great importance. Try to get a clear idea of how these principles are applied in boost (capable of voltage increase), buck (capable of voltage decrease) and cuk (capable of inverting the voltage priority) converters. For example, ripple approximation greatly simplifies the analysis, especially in a well-designed converter where the switching ripples in inductor current and capacitor voltage are comparatively small with respect to the DC component.

Another important element is circuit modeling. You are expected to understand how the DC transformer model is manipulated and solved using conventional circuit analysis

In the IT/ITES segment, with energy efficiency and reduction of operational costs becoming critical business goals, two major trends in data centre infrastructure design are notable today: power efficiency and power distribution.”



—Pramod Agashe, COO of APW

techniques. How the models can be refined to account for loss elements such as inductor-winding resistance and semiconductor 'on' resistance and voltage drops?

Realise the field

Now that you are aware of the basic elements, it's time to realise the *modus operandi* of the actual devices. Switch realisation would be a good starting point. Knowledge about majority carrier devices like MOSFETs and minority carrier devices like BJTs, IGBTs and thyristors is must. Additional knowledge like the operation of unidirectional switches to handle discontinuous conduction mode, may give you an edge over your competitors.

Converter circuit is another important matter that needs thorough understanding. Basically, circuit manipulation, transformer isolation along with circuit evaluation and design related skill sets are in high demand. Try to get familiar with converter dynamics and control. The basic understanding of modelling and averaging approaches will be an added advantage.

In the next step, you have to perform the actual design job by applying all those techniques and concepts discussed so far. Controller design and input filter design are the two major challenges in this field.

Your next challenge is to provide a solution for real-life systems. Go beyond your textbook and enter your own power electronics design laboratory. Try to collect some problems pertaining

to AC and DC equivalent circuit modelling and resonant conversion and chalk out possible solutions. Last but not the least, update yourself on the latest technological advances in this field.

Prepare for the future challenges

Bear in mind that technological advances are meant to increase energy efficiency and reduce operational cost. As a result, the present era is solely dedicated to green energy and energy-saving equipment. Understand this hidden demand and keep pace with it. Your final-year project may be on the utilisation of solar energy or wind energy to run a traditional power system—a solar power fuel dispenser, for example. In fact, there are many ingenious ways to improve power efficiency through various innovations.

According to industry veterans like Pramod Agashe, COO of APW, traditional power distribution units often fail to cope with today's high-density computing environments, so designing of efficient power distribution system for data centres may be a good challenge. Any of these real problems may be your next food for thought.

Perhaps, at this moment the dilemma about your career prospect is driving you nuts. Boost up your brain. Dig down the power field. It will provide you not only a secure job but the opportunity to be an active part of a nation-building story. ●

The author is a research analyst cum journalist at EYF